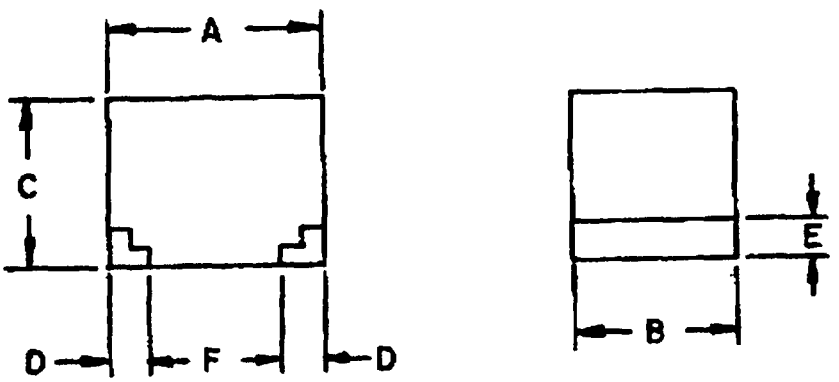
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PERFORMANCE SPECIFICATION SHEET

COILS, CHIP, RADIO FREQUENCY, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the coils described herein shall consist of this document and the latest issue of Specification MIL-C-83446.



	INCHES	MM
A.	.160 Max	4.06 Max
B.	.125 Max	3.18 Max
C.	.125 ±.005	3.18 ±.13
D.	.020 Min	.51 Min
E.	.025 ±.005	.64 ±.13
F.	.090 ±.005	2.29 ±.13

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Fixed, thick film, chip coil.

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REQUIREMENTS

Dimensions and configuration: See figure 1.

Case: Epoxy.

Weight: 0.5 gram maximum.

Operating temperature range: -55° to +125°C.

Storage temperature range: -55° to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): Shall be in accordance with table I.

TABLE I. Electrical characteristics.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test Frequency 1/	Self-resonant frequency (min)	DC resistance (max)	Rated Current (max) 2/	Incremental current 3/
	µH			MHz	MHz	Ohms	mA	mA
01	0.010	36	60	200	1200	0.020	750	
02	0.012	30				0.025		
03	0.015	30				0.035		
04	0.018	30				0.040		
05	0.022	45			1000	0.045		
06	0.027	30				0.050		
07	0.033					0.055	640	
08	0.039		55			0.060	600	
09	0.047			100		0.065	550	
10	0.056					0.060	520	
11	0.068	20			800	0.060	480	
12	0.082	20			800	0.060	470	
13	0.100	30	50	25.0	750	0.160	460	
14	0.120	32			700	0.170	455	
15	0.150	40			650	0.190	450	
16	0.180	30			600	0.200	430	
17	0.220	42			570	0.300	350	
18	0.270	40			500	0.350	310	
19	0.330	36			400	0.720	280	
20	0.390	35			380	0.800	250	
21	0.470	38			450	1.26	230	
22	0.560	40			400	1.30	220	
23	0.680	40			300	1.40	210	
24	0.820	40	45		250	1.50	200	
25	1.0	37			220	1.60	200	
26	1.2	35		7.90	175	1.70	200	
27	1.5	31			135	2.50	190	
28	1.8	30			120	3.00	185	
29	2.2	40			95	3.30	180	

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test Frequency 1/	Self-resonant frequency (min)	DC resistance (max)	Rated current (max) 2/	Incremental current 3/
	µH			MHz	MHz	Ohms	mA	mA
30	2.7	32	45	7.90	85	3.65	175	
31	3.3	32	50		70	4.00	170	
32	3.9	32	50		60	2.10	165	
33	4.7	32	45		50	2.30	165	
34	5.6	35	45		40	2.30	165	
35	6.8	43	50		30	1.30	165	
36	8.2	40	45		22	4.00	120	
37	10.0	40	45		20	2.90	105	58
38	12.0	35	45	2.50	15	3.50	91	53
39	15.0		40		13	2.30	87	50
40	18.0		45		12	3.70	81	46
41	22.0				10	4.00	77	37
42	27	38			9.0	5.00	73	34
43	33	35			9.0	5.70	69	32
44	39				6.5	6.50	65	29
45	47				6.2	6.00	63	29
46	56		40		6.0	6.30	60	28
47	68	30	45		5.8	6.70	57	26
48	82	35	45		5.2	7.50	55	23
49	100	33	40		4.0	13.0	50	20
50	120	33		0.790	4.0	14.0	40	20
51	150	35			3.9	16.0	36	20
52	180				3.5	18.0	33	19
53	220				3.2	21.0	28	18
54	270				3.1	23.0	25	17
55	330				2.8	37.0	20	16
56	390				2.5	39.0	17	16
57	470				2.3	45.0	15	15
58	560				2.2	47.0	13	14
59	680				2.1	62.0	12	13
60	820				2.1	78.0	11	11
61	1000	33			1.1	96.0	10	10

1/ Test frequency range: 0.25 through 25 MHz.

Inductance shall be measured on instruments such as the HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with TF-405 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.038 µH) and residual Q-Meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

Test frequency range: 25 through 200 MHz.

Inductance shall be measured on instruments such as the HP190-A Q Meter, HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with TF-405 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

2/ Maximum current allowed not to exceed specified temperature rise.

3/ Applicable to ferrite core only.

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Electrical characteristics (final): See table II.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	---	---	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

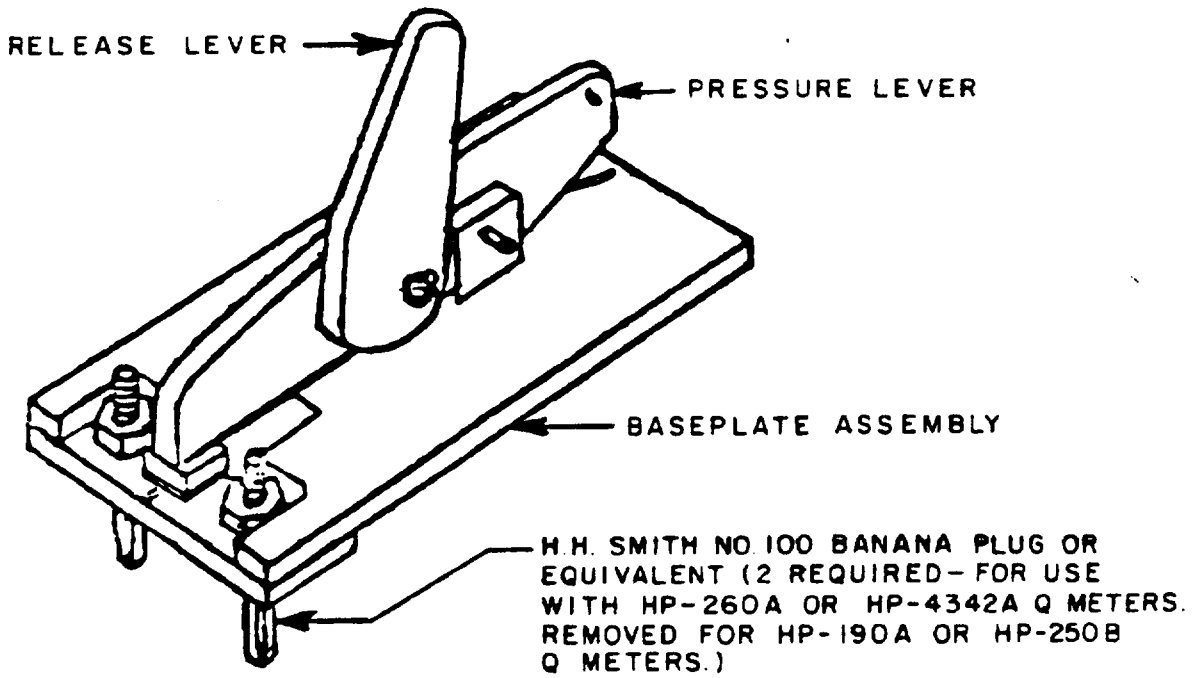


FIGURE 2. Typical chip coil test fixture (TF-405, or equal).

Part number. The part number shall be in the following form.

M83446/14- 04 B

Sequentially assigned dash numbers (see table I)

Termination finish (see MIL-C-83446)

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Navy - EC
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User activities:
Navy - OS
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Preparing activity:
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